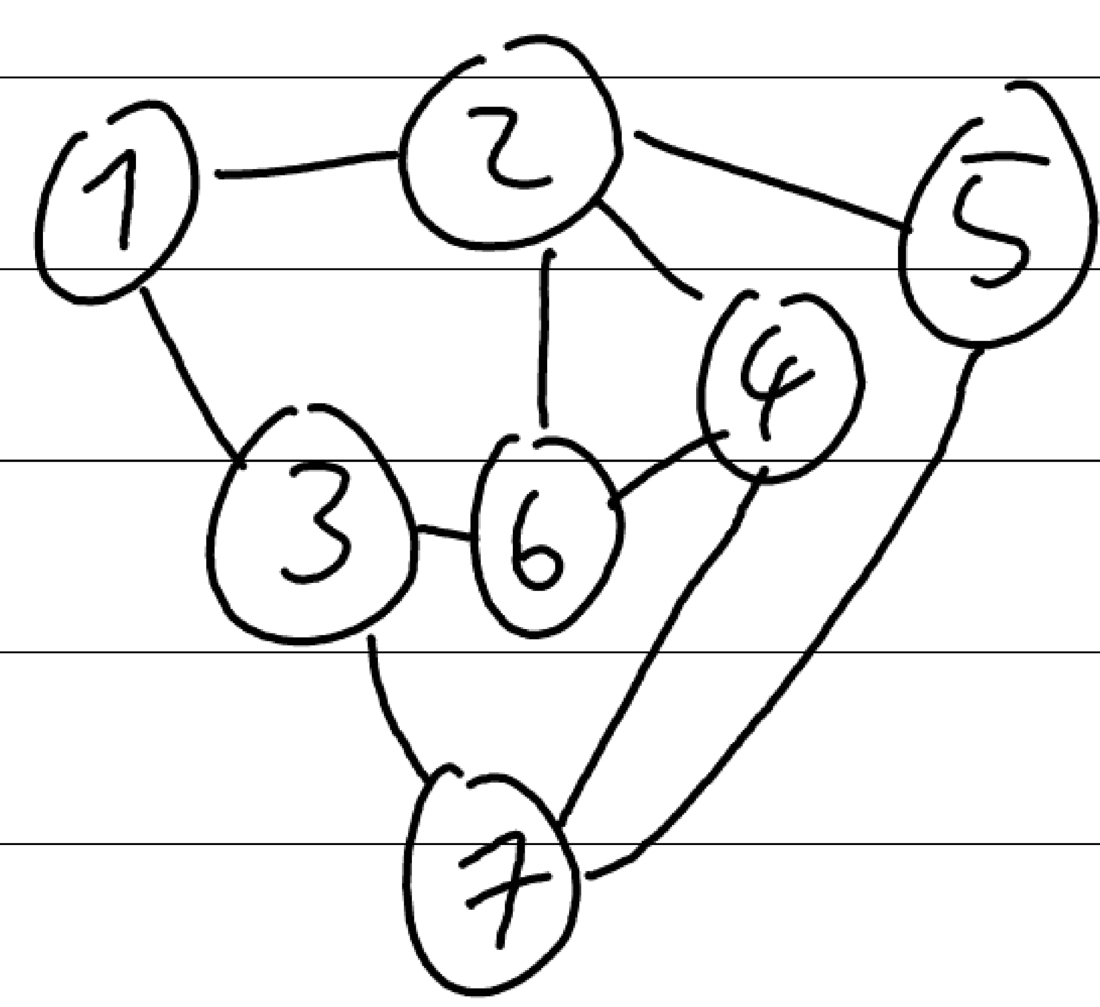
13 Graph neural networks

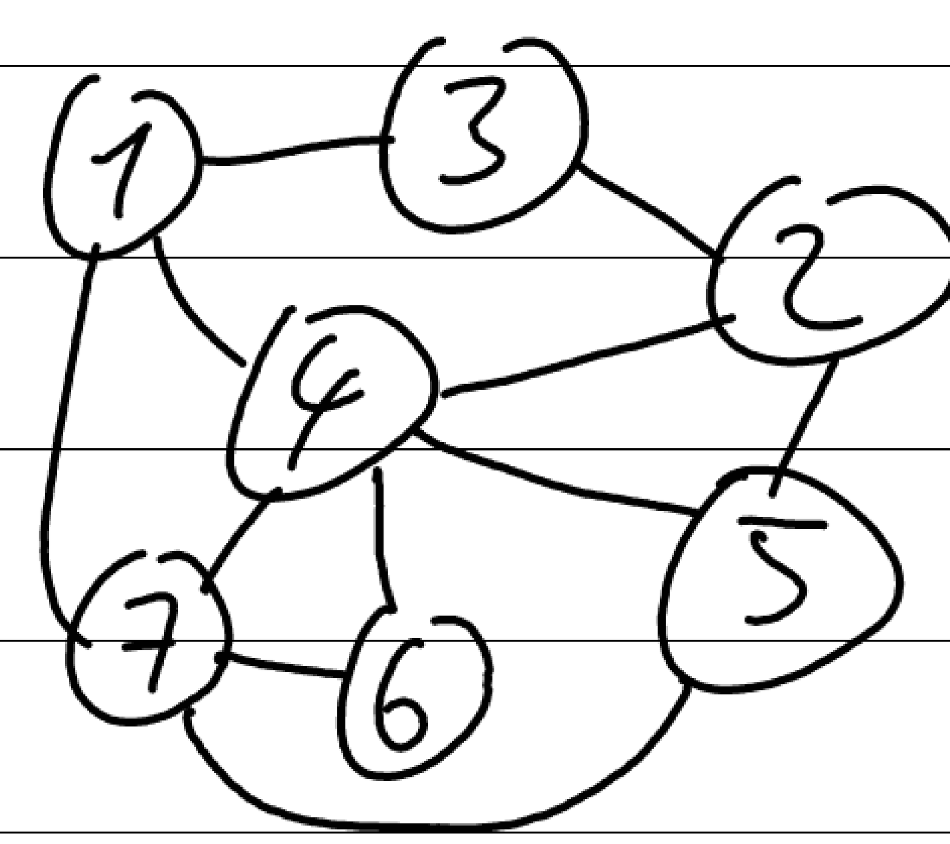
13.1

a) $A = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$

b) $A = \left(\begin{array}{cccccccc} 0 & & & & & & & \\ 0 & 0 & & & & & & \\ 0 & 1 & 0 & & & & & \\ 1 & 0 & 0 & 0 & & & & \\ 0 & 1 & 0 & 0 & 0 & & & \\ 0 & 1 & 0 & 0 & 1 & 0 & & \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \end{array} \right.$ ... symmetric ...

13.2

1: 

2: 

13.3

(i) three steps: a) 9
b) 3

(ii) seven steps: a) 879
b) 203

13.4 Position (m,m) of A^2 contains the number of walks of length two from node m to m, so exactly 'walking back and forth' to every connected node.

Permutation matrix P: exactly one '1' in each row and column, e.g., $\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}a&b\\c&d\end{pmatrix}=\begin{pmatrix}c&d\\a&b\end{pmatrix}$, with $P^TP=I$,

$$\begin{pmatrix}c&d\\a&b\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix}=\begin{pmatrix}d&c\\b&a\end{pmatrix} \text{ but } \begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix}=\begin{pmatrix}1&0\\0&1\end{pmatrix}$$

13.5 $1 \to 3$
$2 \to 1$
$3 \to 6$
$4 \to 2$
$5 \to 4$
$6 \to 5$

$$\Rightarrow P=\begin{pmatrix}0&0&1&0&0&0\\1&0&0&0&0&0\\0&0&0&0&0&1\\0&1&0&0&0&0\\0&0&0&1&0&0\\0&0&0&0&1&0\end{pmatrix}$$

13.6

to $\mathrm{sig}[\beta_4+w_4 H_4 \mathbf{1}]=\mathrm{sig}[\beta_4+w_4H_4P\mathbf{1}]$, $\mathbf{1}_{N\times 1}$

e.g., $\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}1\\1\end{pmatrix}$

$(P\mathbf{1})_i=\sum_{j=1}^{N}P_{ij}\cdot 1 = 1$ as P_{ij} is 1 exactly once

Thus $P\mathbf{1}=\mathbf{1}$ and the rest follows $\square$

13.7 $H_{k+1} = \text{GraphLayer}[H_k, A]$

$$= a\left[\beta_k 1^T + \Omega_k \begin{pmatrix} H_k \\ H_k A \end{pmatrix}\right]$$

zz $\text{GraphLayer}[H_k, A]P = \text{GraphLayer}[H_k P, P^T A P]$

$$\text{GraphLayer}[H_k, A]P = a\left[\beta_k 1^T + \Omega_k \begin{pmatrix} H_k \\ H_k A \end{pmatrix}\right]P$$

$$= a\left[\beta_k P + 1^T P + \underset{\parallel}{\Omega_k} \begin{pmatrix} H_k \\ H_k A \end{pmatrix} P\right]$$

$$= a\left[\beta_k P + 1^T P + \underbrace{(\Omega'_k \ \Psi_k) \begin{pmatrix} H_k \\ H_k A \end{pmatrix} P}_{\Omega'_k H_k P + \Psi_k H_k A P}\right]$$

$$= \Omega'_k H_k P + \Psi_k H_k P P^T A P$$

ces $PP^T = I$ $\square$

13.8 degree matrix D: $D_{ij} = |\text{neighbors}|$ für $i=j$, 0 für $i \neq j$

$$D_a = \begin{pmatrix} 5 &&&&&& \\ & 4 &&&&& \\ && 2 &&&& \\ &&& 3 &&& \\ &&&& 2 && \\ &&&&& 2 & \\ &&&&&& 2 \end{pmatrix} \quad D_b = \begin{pmatrix} 3 &&&&&&& \\ & 3 &&&&&& \\ && 2 &&&&& \\ &&& 2 &&&& \\ &&&& 3 &&& \\ &&&&& 3 && \\ &&&&&& 3 & \\ &&&&&&& 3 \end{pmatrix}$$

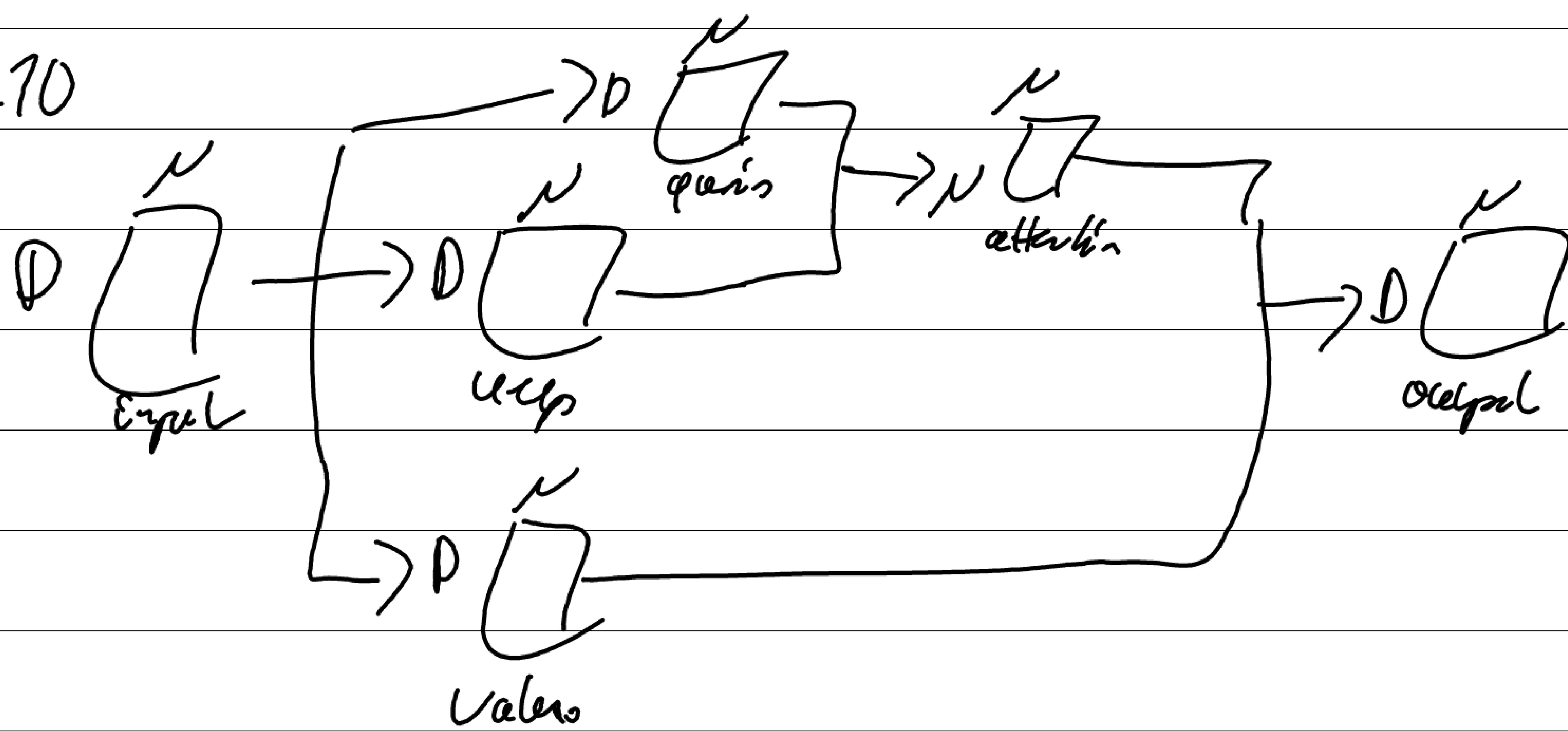
13.9 $\quad agg[n] = \frac{1}{1+|ne[n]|}\left(h_n + \sum_{m \in ne[n]} h_m\right)$

As $\quad h_n + \sum_{m \in ne[n]} h_m = H(I+A) \quad \left(\text{as } HA = \sum_{m \in ne[n]} h_m\right),$

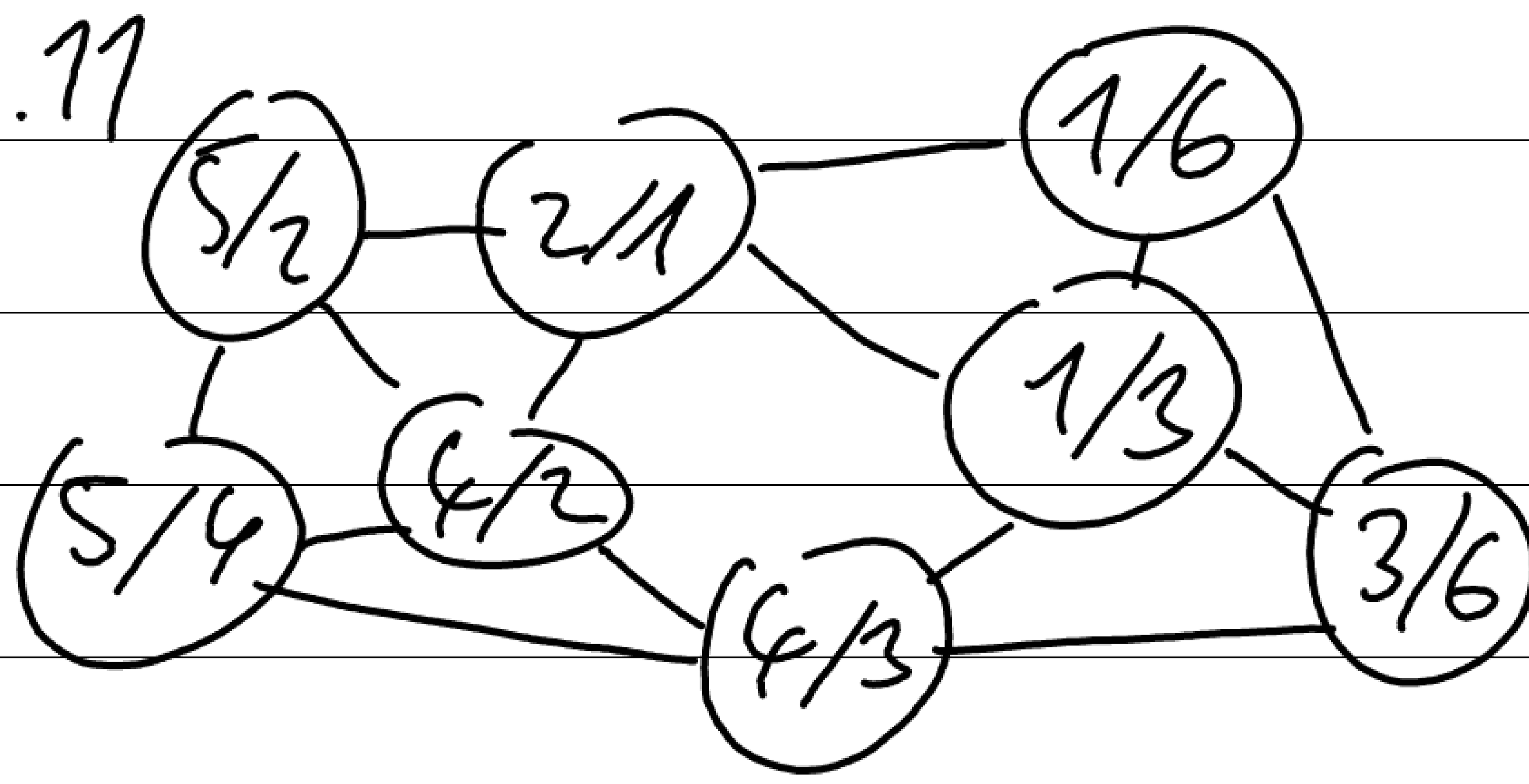
and $\quad \frac{1}{I+|ne[n]|} = (I+D)^{-1},$

$$agg[n] = H(I+A)\,(I+D)^{-1} \qquad \square$$

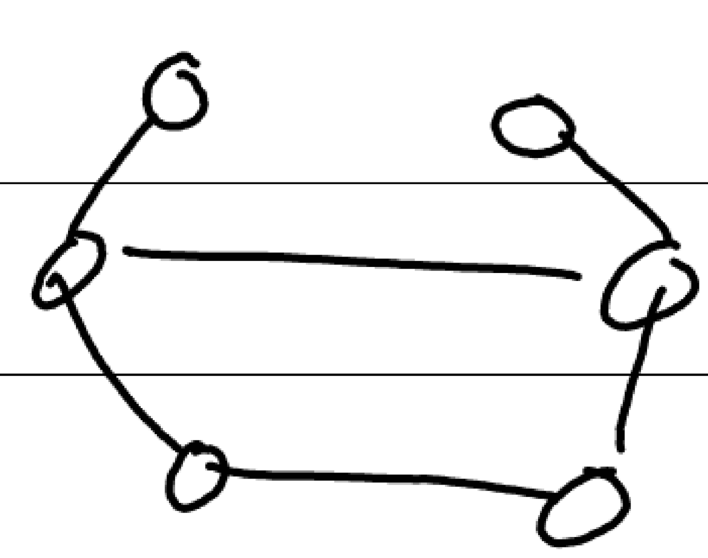
13.10



13.11



13.12



13.13 A_{node} and A_{edge} are not directly expressible from one another alone, but come from the same incidence matrix.

13.14

$$H_{k+1} = a\left[\beta_k 1^T + \lambda_k H_k (A+I) + \phi_k E_k N\right]$$

with E: edge embeddings and N: indices of edges that connect to node k in column k